

DIABETES PREDICTION

Using Machine Learning in Weka

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Presentation Outline

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Introduction

Why does diabetes prediction matter?

537M+

Adults with Diabetes Globally
(IDF, 2021)

780M

Projected cases by 2045
(growth of ~45%)

**Early
Prediction**

Reduces complications
by up to 40%

- ▶ Diabetes is a chronic metabolic disorder requiring early intervention for effective management.
- ▶ Machine learning enables pattern recognition in clinical data to predict diabetes risk before onset.
- ▶ Weka provides a no-code GUI for implementing and comparing ML classifiers on biomedical datasets.
- ▶ This project predicts diabetes using 8 clinical features from the Pima Indians dataset.

Dataset: Pima Indians Diabetes

768 patient records | 8 clinical features | Binary classification

Pregnancies

Times pregnant

Glucose

Plasma glucose (OGTT)

Blood Pressure

Diastolic BP (mm Hg)

Skin Thickness

Triceps skinfold (mm)

Insulin

2-hr serum insulin

BMI

Body mass index

Diabetes Pedigree

Family risk score

Age

Patient age (years)

Target Variable: Outcome

Non-Diabetic (0): 500 instances — 65.1%

Diabetic (1): 268 instances — 34.9%

Methodology: Weka Pipeline

Step-by-step data mining workflow in Weka Explorer

1

Load
Dataset

Import diabetes.csv into
Weka Explorer

2

Handle
Missing Values

Apply
ReplaceMissingValues
filter

3

Convert
Class Type

NumericToNominal on
Outcome column

4

Set Class
Attribute

Mark Outcome as target
variable

5

Train
Classifiers

J48, Random Forest,
Naive Bayes, Logistic

6

Evaluate
Models

10-fold Cross-Validation

7

Compare
& Select

Accuracy, F1, AUC
comparison

8

Predict
New Data

Apply saved model to
unseen records

Data Preprocessing in Weka

Ensuring clean, model-ready data

Step 1: Load & Inspect

- Open Weka → Explorer → Preprocess → Open file
- Load diabetes.csv (768 instances, 9 attributes)
- Review attribute types and value distributions
- Check for missing value indicators in summary

Step 3: Convert Numeric → Nominal

- J48 requires nominal (categorical) class attribute
- Filter: Unsupervised → Attribute → NumericToNominal
- Apply to Outcome column only
- Result: 0 = Non-diabetic | 1 = Diabetic (as categories)

Step 2: Replace Missing Values

- Filter: Unsupervised → Attribute → ReplaceMissingValues
- Columns with 0s as proxies: Glucose, BMI, Insulin, SkinThickness, BP
- Filter replaces invalid zeros with column mean
- Verify: no '?' symbols remain in attribute table

Step 4: Set Class Attribute

- In Preprocess tab, select Outcome in attribute list
- Set as class attribute using dropdown menu
- Weka highlights class column distinctly
- Data is now ready for Classify tab

Classification Models

Four ML algorithms applied in Weka Classify tab

J48 Decision Tree

Weka Path

trees → J48

Strength

Human-interpretable tree structure

Limitation

Susceptible to overfitting

Random Forest

Weka Path

trees → RandomForest

Strength

Highest accuracy; reduces overfitting via ensemble

Limitation

Less interpretable than single tree

Naive Bayes

Weka Path

bayes → NaiveBayes

Strength

Fast, probabilistic; works well with small data

Limitation

Independence assumption often violated

Logistic Regression

Weka Path

functions → Logistic

Strength

Strong AUC; interpretable coefficients

Limitation

Assumes linear decision boundary

All models evaluated using 10-fold Cross-Validation for reliable, unbiased accuracy estimation.

Model Performance Results

10-fold Cross-Validation across all four classifiers

BEST MODEL

82–85%

Random Forest

F1: 0.79 AUC: 0.89

77–80%

Logistic Regression

F1: 0.76 AUC: 0.83

76–78%

J48 Decision Tree

F1: 0.73 AUC: 0.81

74–76%

Naive Bayes

F1: 0.72 AUC: 0.82

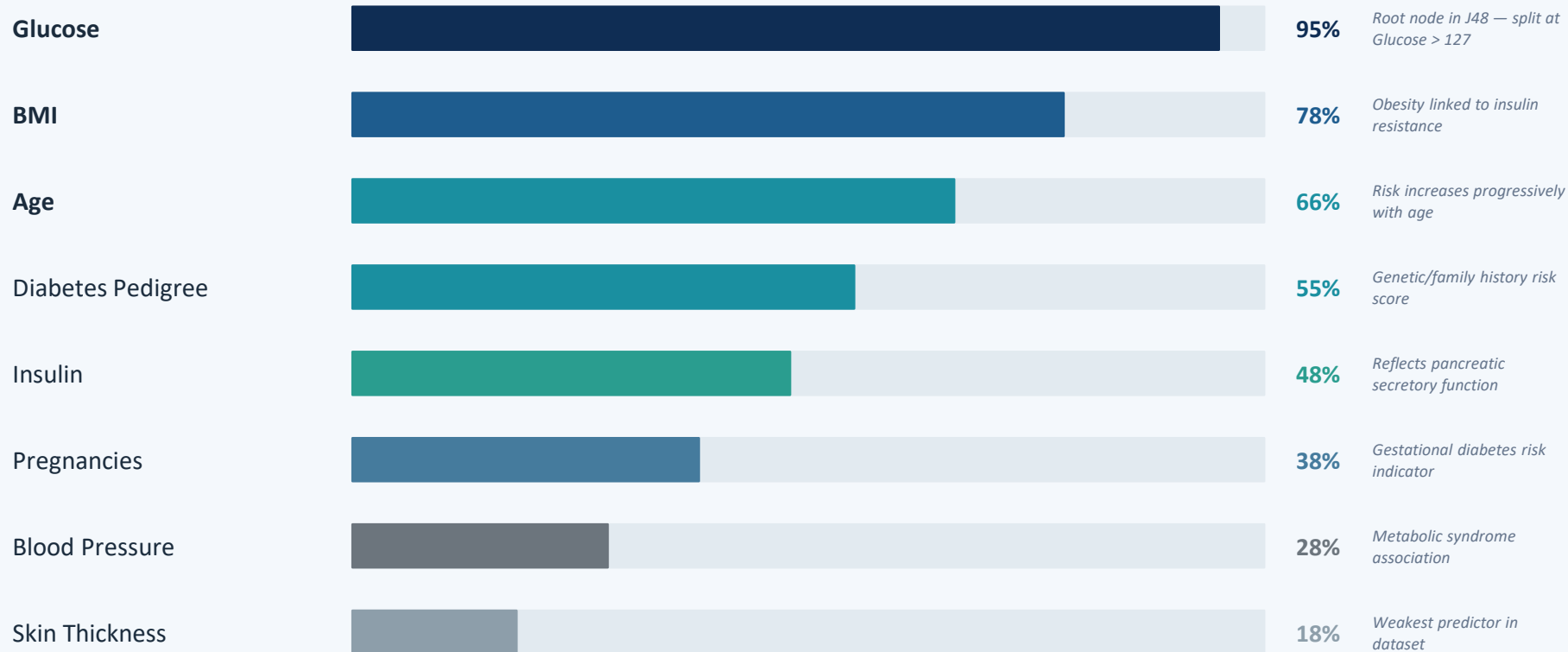
Sample Confusion Matrix (J48)		
	Pred: 0	Pred: 1
Actual: 0	TN = 350	FP = 50
Actual: 1	FN = 30	TP = 138

Key Insights

- ✓ Random Forest provides best accuracy with lowest overfitting
- ✓ J48 offers highest interpretability for clinical use
- ✓ False Negatives (FN) are clinically critical — missed diabetics
- ✓ AUC > 0.80 for all models — strong discrimination ability

Feature Importance Analysis

Which clinical factors predict diabetes most strongly?



Importance scores derived from J48 decision tree structure and Random Forest attribute evaluation.

Predicting New Patient Data

Applying the trained model to unseen records

Step 1

Prepare CSV with same 8 attributes as training data (no Outcome column)

Step 2

Load saved model: Classify → More options → Load Model

Step 3

Select Supplied test set → browse to new patient CSV

Step 4

Click Start → Weka outputs predicted Outcome for each patient

Example Predictions

Glucose	BMI	Age	Blood Pressure	Prediction	Confidence
120	28.0	45	80	Non-Diabetic	High
160	35.4	50	88	Diabetic	High
95	22.1	30	70	Non-Diabetic	Medium
185	40.2	55	92	Diabetic	High

Conclusions & Future Scope

- ✓ Random Forest achieved best accuracy (82–85%) with strong AUC (~0.89)
- ✓ J48 provides the most clinically interpretable model via decision tree visualization
- ✓ Glucose level is the single most important predictor — split at Glucose > 127
- ✓ BMI and Age follow as second and third most important clinical risk factors
- ✓ Weka is a powerful, no-code platform for biomedical ML tasks
- ✓ The pipeline demonstrates effective early-risk stratification from clinical data

Future Scope

- Apply deep learning models (LSTM, Transformers) for improved accuracy
- Extend to diverse populations and multi-class diabetes types
- Deploy as a real-time clinical decision-support web tool
- Integrate EHR data for continuous, personalized risk monitoring

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